



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Mid SOLUTION	Marks 25 Weight 25%	Mar 09, 2026
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Important Notes

Clearly write your name and student ID on top.

CLO-1	Apply knowledge of signals and systems to categorize and perform basic operations on signals and systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)
CLO-2	Calculate response of DT and CT LTI systems to various inputs and categorize LTI systems.	PLO-1 (Engineering Knowledge)	Cognitive Level-3 (Applying)

Note: $u(t)$, where used, represents the unit step function.

Problem 1 [CLO1] [1+2+2 Points]

Consider α to be a real finite constant. Answer the following for each case: for what value[s] of α is the signal periodic? Use that value and find the period.

- (a) $x(t) = \alpha + 7 \sin(3t)$
- (b) $x(t) = \sin(8t + 30^\circ) + \alpha u(t)$
- (c) $x(t) = e^{(1-\alpha)t} e^{j2t}$

Solution:

(a)

Adding a finite real constant does not affect the periodicity of a signal (can be verified graphically or mathematically), so if a signal $y(t)$ is periodic, so is $\alpha + y(t)$. Since $7 \sin(3t)$ is periodic, so is $\alpha + 7 \sin(3t)$ for all finite real constants α . Period of the signal is $T_0 = \frac{2\pi}{3}$.

(b)

Since $u(t)$ is not periodic, given signal may only be periodic for $\alpha = 0$. At this value we have

$$\text{(b) } \sin(8(t + \frac{2\pi}{8}) + 30) = \sin(8t + 2\pi + 30) = \sin(8t + 30). \\ \omega_0 = 8 \text{ and } T_0 = \frac{2\pi}{8} = \frac{\pi}{4}.$$

(c)

Since $e^{(1-\alpha)t}$ is not periodic in general (e.g., check graphically), the given signal may only be periodic for $\alpha = 1$. At this value we have $x(t) = e^{j2t}$ and

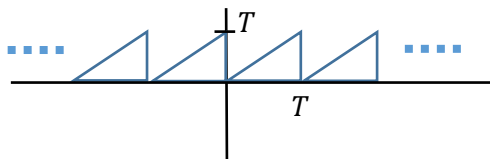
(c) $e^{jt} = \cos(t) + j \sin(t)$ is periodic with fundamental period 2π , so e^{j2t} is periodic with fundamental period $\frac{2\pi}{2} = \pi$, and fundamental frequency $\omega_0 = 2$.

Problem 2 [CLO1] [2.5+2.5 Points]

Find energy and power of the following signals

(a) $x(t) = \begin{cases} \sqrt{t}, & 0 \leq t \leq 10 \\ 0, & \text{otherwise} \end{cases}$

(b) $x(t)$ is as a periodic function as shown below, with $T = 1$.



Solution:

(a) $E_x = \int_0^{10} |\sqrt{t}|^2 dt = \int_0^{10} t dt = \left. \frac{t^2}{2} \right|_0^{10} = \frac{100}{2} = 50$

$P_x = 0$ (from formula of power for aperiodic signals, it is easy to see that we will have $\frac{50}{\infty} = 0$)

(b) Easy to see that the total absolute squared area under the periodic signal will be infinite, i.e., $E_x = \infty$

Next, using formula of power for a periodic signal, and noticing that in the period $[0, T]$ the signal can be written as $x(t) = t$, we have

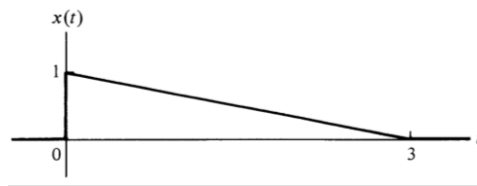
$$P_x = \frac{1}{T} \int_0^T |t|^2 dt = \left. \frac{t^3}{3} \right|_0^T = \frac{1}{3}$$

Where we have inserted $T = 1$, as given.

Problem 3 [CLO1] [3+2 Points]

Plot the signals as directed in each case

(a) Given

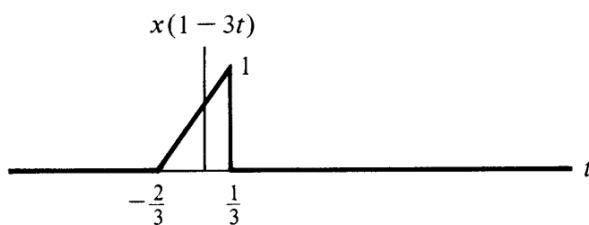
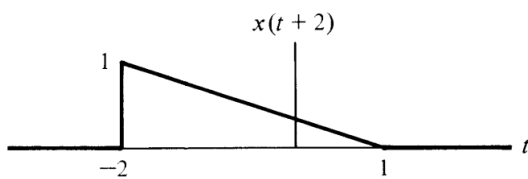
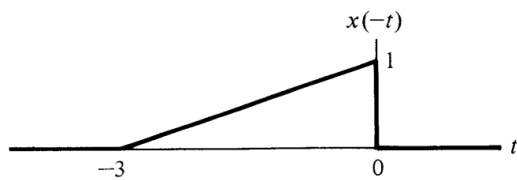


Plot $x(-t)$, $x(t+2)$, and $x(1-3t)$

(b) Plot $y(t) = \int_{-\infty}^{\infty} t^2 \delta(2t-4) dt$

Solution:

(a)



$$(b) y(t) = \int_{-\infty}^{\infty} t^2 \delta(2t-4) dt = \int_{-\infty}^{\infty} 2^2 \delta(2t-4) dt = 4 \int_{-\infty}^{\infty} \delta(2t-4) dt = 4 \int_{-\infty}^{\infty} \delta(v) \frac{dv}{2} = 2$$

The plot is basically a horizontal line of magnitude 2.

Problem 4 [CLO2] [2.5+2.5 Points]

Determine whether each of the following systems is (i) Linear (ii) Time Invariant. Provide mathematical justifications for each. Note that, in each case $x(t)$ is the input and $y(t)$ is the output.

(a) $y(t) = 4^k x(t) + t^4 x(t)$, where $0 < k < 4$

(b) $y(t) = 4x(t) + \cos(4) \cos(x(t))$

Solution

(a) In the given case, the system is $y(t) = S\{x(t)\} = 4^k x(t) + t^4 x(t)$. For linearity, we must show that

$$S\{\alpha_1 x_1(t) + \alpha_2 x_2(t)\} = \alpha_1 S\{x_1(t)\} + \alpha_2 S\{x_2(t)\}$$

Now,

$$LHS = 4^k (\alpha_1 x_1(t) + \alpha_2 x_2(t)) + t^4 (\alpha_1 x_1(t) + \alpha_2 x_2(t))$$

And

$$RHS = \alpha_1 (4^k x_1(t) + t^4 x_1(t)) + \alpha_2 (4^k x_2(t) + t^4 x_2(t))$$

Simple algebraic manipulation shows that $LHS = RHS$, leading to the conclusion that the system is linear.

Next, for time-invariance, we have to show that $S\{x(t - T)\} = y(t - T)$, this would fail since time variable t appears outside of $x(t)$ as well leading to the conclusion that the system is time-varying.

(b) In the given case, the system is $y(t) = S\{x(t)\} = 4x(t) + \cos(4) \cos(x(t))$. For linearity, we must show that

$$S\{\alpha_1 x_1(t) + \alpha_2 x_2(t)\} = \alpha_1 S\{x_1(t)\} + \alpha_2 S\{x_2(t)\}$$

It is easy to see that since $\cos(a + b)$ is, in general, not equal to $\cos(a) + \cos(b)$, the above relation will not hold. Leading to the conclusion that the system is nonlinear.

Next, for time-invariance, we have to show that $S\{x(t - T)\} = y(t - T)$. For this, we have

$$LHS = 4x(t - T) + \cos(4) \cos(x(t - T))$$

And

$$RHS = y(t - T) = 4x(t - T) + \cos(4) \cos(x(t - T))$$

Since $LHS = RHS$, we conclude that the system is Time Invariant.

Problem 5 [CLO2] [1+1+3 Points]

The impulse response of a DT LTI system is given as

$$h[n] = \begin{cases} 1, & n = 0,1,2,3 \\ -1, & n = 4,5 \\ 0, & \text{otherwise} \end{cases}$$

- (a) Use $h[n]$ to check if the system is memoryless (justify your answer)
- (b) Use $h[n]$ to check if the system is BIBO stable (justify your answer)
- (c) Find and plot the output of the system when the input is

$$x[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 2 \\ 0, & \text{otherwise} \end{cases}$$

Solution:

- (a) The system has memory since $h[n] \neq \alpha \delta[n]$
- (b) The system is BIBO stable since $\sum |h[n]| = 6 < \infty$
- (c) Since system is LTI, we have $y[n] = x[n] * h[n]$. The convolution can be solved in multiple ways (analytically, graphically etc.), leading to the output

$$y[n] = \begin{cases} 2, & n = 0,1 \\ 3, & n = 2,3 \\ -1, & n = 4,5,6,7 \\ 0, & \text{otherwise} \end{cases}$$

Which is easy to plot.